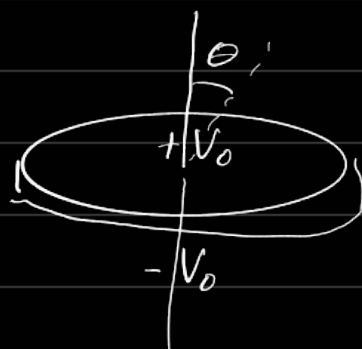


$$\nabla^2 V(\vec{r}) = 0$$

$$V(R, \theta, \varphi) = \begin{cases} V_0 & 0 \leq \theta \leq \pi/2 \\ -V_0 & -\pi/2 \leq \theta \leq \pi \end{cases}$$



$$\bullet \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} - \frac{\lambda^2}{r^2} R = 0 \quad 0 \leq r \leq R$$

$$\bullet -\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta}{d\theta} \right] = \lambda^2 \Theta \quad 0 \leq \theta \leq \pi$$

Legendre equation: $(1-x^2)y'' - 2xy' + \lambda y = 0$

which can be rewritten as:

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \lambda y = 0$$

Note this is a Sturm-Liouville problem:

$$-\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] = \lambda y$$

Define the operator

$$\mathcal{L}_p := -\frac{d}{dx} \left[(1-x^2) \frac{d}{dx} \right]$$

which acts in an "odd" way:

$$\mathcal{L}_p f = -\frac{d}{dx} \left[(1-x^2) \frac{df}{dx} \right]$$

• Let's convert this spherical:

$$\cos(\theta) = x$$

$$\frac{d}{dx} = \frac{d\theta}{dx} \frac{d}{d\theta}$$

$$\rightarrow \frac{d}{dx} (\cos \theta) = \frac{d}{dx} (x)$$

$$-\sin \theta \frac{d\theta}{dx} = 1 \Rightarrow \frac{d\theta}{dx} = -\frac{1}{\sin \theta}$$

$$\text{Thus } \mathcal{L}_p f = - \left(-\frac{1}{\sin \theta} \frac{d}{d\theta} \right) \left[\frac{\sin^2 \theta}{-\sin \theta} \frac{df}{d\theta} \right]$$

$$= -\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{df}{d\theta} \right]$$

$$r^2 R'' + 2r R' - \lambda^2 R = 0$$

$$r = e^w \quad d = dw$$

$$\frac{d}{dr} = \frac{d}{dr} \frac{d}{dw}$$

$$1 = e^w \frac{dw}{dr} \quad \frac{d}{dr} = e^{-w} \frac{d}{dw}$$

$$e^{-w} \frac{d}{dw} \left[e^{-w} \frac{d}{dw} \right] = e^{-w} \left[(-1) e^{-w} \frac{d}{dw} + e^{-w} \frac{d^2}{dw^2} \right]$$

$$\frac{d^2}{dr^2} = e^{-2w} \left[\frac{d^2}{dw^2} - \frac{d}{dw} \right]$$

$$\left(\frac{d^2}{dw^2} - \frac{d}{dw} \right) R + 2e^w \left(e^{-w} \frac{d}{dw} R \right) = \lambda^2 R = 0$$

$$\frac{d^2 R}{dw^2} - R' + 2R' - \lambda^2 R = 0$$

$$R'' + R' - \lambda^2 R = 0$$

$$R(w) = e^{wa}$$

$$-a^2 + a - \lambda^2 = 0$$

$$a^2 + a - \lambda^2 = 0$$

$$\frac{-1 \pm \sqrt{1+4\lambda^2}}{2}$$

$$R(r) = e^{\ln(r)a} = r^a$$

$$R(r) = A r^a + B r^{-a}$$

Note that $\lim_{r \rightarrow 0} r^{-a}$ blows up

$$\text{and so } R(r) = A r^a$$

$$\frac{d}{dr} (r^a) = \ln(a) r^a$$

$$(\ln(a))^2 r^a + \frac{2}{r} \ln(a) r^a - \frac{\lambda^2}{r^2} r^a = 0$$

$$(\ln(a))^2 r^a + 2\ln(a) r^{a-1} - \lambda^2 r^{a-2} = 0$$

